

CPR Esoft Quiz 3

Question #: 1

A 48-year-old man comes to the physician because of a 2-week history of red spots on his skin. He reports easy bruising but denies fever, recent infections, or new medications. Physical examination shows multiple petechiae over the arms and legs. Laboratory studies show a platelet count of 20,000/ μ L (normal: 150,000–450,000/ μ L). Hemoglobin and white blood cell counts are within normal limits. Which of the following sets of laboratory findings is most likely to be observed in this patient?

- A. I
- B. II
- C. III
- D. IV
- E. V

Attachment:

	Bleeding time (BT)	Activated partial thromboplastin time (APTT)	Prothrombin time (PT or INR)
I	Increased	Increased	Normal
II	Normal	Increased	Normal
III	Increased	Normal	Normal
IV	Normal	Normal	Increased
V	Increased	Increased	Increased

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Question #: 2

A 35-year-old woman comes to the physician because of a 2-week history of easy bruising and intermittent abdominal pain. She reports no history of alcohol use. Physical examination is notable for mild jaundice. Abdominal ultrasonography shows stones in the common bile duct. Laboratory studies show:

- Elevated serum alkaline phosphatase (ALP)
- Elevated gamma-glutamyl transpeptidase (GGT)
- Prolonged prothrombin time (PT)

A diagnosis of vitamin deficiency is made. Which of the following enzyme activities is most likely decreased in this patient as a result of this deficiency?

- A. γ -glutamyl carboxylase in the liver
 - B. 25-hydroxylase in the liver
 - C. 1-hydroxylase in the kidney
 - D. Prolyl hydroxylase
 - E. Superoxide dismutase
-

Question #: 3

A 10-month-old boy is brought to the physician by his mother because of a 3-day history of swelling of the scalp. He also has a history of prolonged bleeding from the umbilical stump at birth. Physical examination shows swelling of the scalp and multiple bruises on his arms and legs in various stages of healing. Laboratory studies show:

Prolonged activated partial thromboplastin time (aPTT),

Normal prothrombin time (PT/INR),

Normal platelet count, bleeding time, and platelet aggregation.

Which of the following is the most likely diagnosis?

- A. Hemophilia A
 - B. Von Willebrand disease
 - C. Bernard-Soulier syndrome
 - D. Vitamin K deficiency
 - E. Scurvy
 - F. Thrombocytopenia
-

Question #: 4

A 60-year-old man is brought to the emergency department because of a 1-hour history of progressive shortness of breath, chest pain, and palpitations. Diagnostic studies confirm pulmonary embolism, and he is started on intravenous unfractionated heparin. Activated partial thromboplastin time (aPTT) is monitored to assess the effect of heparin on the intrinsic coagulation pathway. Which of the following are most likely components of this pathway?

- A. GPIb and GPIIb/IIIa
- B. von Willebrand factor and tissue factor
- C. Factors II, VII, IX and X
- D. Factors VIII, IX, XI and XII
- E. Factors III, VII, X and prothrombin
- F. Factors V, X, XIII, prothrombin and fibrinogen

Question #: 5

A 26-year-old woman comes to the physician because of a 2-day history of pain, swelling, and redness in her left forearm. Three days ago, she scraped her arm on a tree branch while gardening. She has no significant medical history. Her temperature is 38.3°C (101°F). Physical examination shows a 4-cm area of erythema, warmth, and tenderness over the mid-forearm with mild edema. A small abrasion with crusted exudate is present at the center of the lesion. Which of the following mediators is most likely responsible for the pain and erythema observed at the site of injury?

- A. Leukotrienes
- B. Prostaglandin E2
- C. Prostacyclin
- D. Thromboxane A2
- E. Arachidonic acid

Question #: 6

A 3-year-old girl is brought to the emergency department by her father because of a sudden onset of dry cough and shortness of breath. Her heart rate and respiratory rate are increased. Physical examination shows an anxious child using accessory muscles of respiration. Auscultation of the lungs shows bilateral expiratory wheezes. An anti-allergic medication is administered, and her symptoms rapidly improve. Which of the following best describes the pathophysiologic mechanism underlying this patient's condition?

- A. Bronchoconstriction is mediated by thromboxane A2
- B. Cyclooxygenase activity in mast cells results in synthesis of leukotrienes
- C. Slow-reacting substance-A mediates bronchodilation in the lungs
- D. Lipoxygenase activity results in the formation of PGI₂
- E. Antigenic interactions in mast cells initiate cysteinyl leukotriene synthesis

Question #: 7

A 25-year-old woman at 28 weeks' gestation is brought to the emergency department because of sudden-onset vaginal bleeding and lower abdominal cramping. Physical examination shows uterine tenderness and signs of early labor. Abdominal ultrasound confirms increased uterine activity. Analysis of uterine tissue shows increased formation of an eicosanoid that stimulates uterine contractions. Which of the following molecules is most likely responsible for this effect on the uterine tissue?

- A. Leukotriene D₄
 - B. Thromboxane A₂
 - C. Leukotriene B₄
 - D. Leukotriene A₄
 - E. Prostaglandin F_{2α}
 - F. Prostacyclin (PGI₂)
-

Question #: 8

A 21-year-old woman is brought to the emergency department because of a laceration on her right hand sustained while cutting vegetables. Shortly after the injury, there is localized vasoconstriction, platelet aggregation, and lymphocyte activation at the wound site. Which of the following eicosanoids is most likely responsible for mediating this localized response?

- A. Thromboxane A₂
 - B. Prostacyclin
 - C. Prostaglandin E₂
 - D. Prostaglandin F_{2α}
 - E. Prostaglandin G₂
 - F. Leukotriene D₄
-

Question #: 9

A 6-year-old boy is brought to the physician by his parents because of a 4-day history of fatigue and pallor. He has been taking a sulfanilamide antibiotic for the past 5 days. Physical examination shows mild jaundice. Laboratory studies show anemia with the presence of bite cells and Heinz bodies on peripheral smear. Genetic testing shows a pathogenic variant in the glucose-6-phosphate dehydrogenase (G6PD) gene. Which of the following family members is most likely to be at risk of developing similar manifestations after exposure to a similar drug?

- A. Father
 - B. Mother
 - C. Maternal uncle
 - D. Paternal aunt
 - E. Elder sister
-

Question #: 10

A 4-year-old boy is brought to the physician by his parents because of a 2-day history of pallor and dark-colored urine. Physical examination shows mild jaundice and splenomegaly. Laboratory studies show anemia with evidence of increased red blood cell destruction. Further testing shows decreased activity of a metabolic pathway that impairs the red blood cell's ability to respond to oxidative stress, resulting in splenic sequestration and hemolysis. Which of the following best describes the main role of this pathway in the affected cells?

- A. Generates reactive oxygen species
 - B. Generates ATP by substrate level phosphorylation
 - C. Generates NADPH for glutathione reductase
 - D. Provides thromboxane A2 for platelet aggregation
 - E. Forms 2,3-bisphosphoglycerate
 - F. Generates hydrogen peroxide
 - G. Generates NADPH for fatty acid synthesis
-

Question #: 11

A 16-month-old boy is brought to the physician by his mother because of a 4-day history of fever and productive cough. His medical history is notable for multiple bacterial and fungal infections over the past 4 months. Physical examination shows hepatosplenomegaly and generalized lymphadenopathy. Laboratory studies confirm impaired intracellular killing by neutrophils. The dihydrorhodamine test is negative, indicating the absence of respiratory burst activity in macrophages. Which of the following reactions is most likely directly affected in this patient?

- A. Formation of hypochlorous acid (HOCl) from H_2O_2
 - B. Oxidation of glucose 6-phosphate to 6-phosphogluconate
 - C. Formation of superoxide from oxygen
 - D. Conversion of H_2O_2 to water
 - E. Formation of nitric oxide from arginine
 - F. Formation of prostaglandins from arachidonic acid
 - G. Conversion of adenosine to inosine
-

Question #: 12

A 30-year-old man presents to his physician for a routine checkup. He has no significant past medical history and reports no symptoms. Laboratory studies show normal red blood cell counts. In vitro testing of his erythrocytes shows that hydrogen peroxide generated during normal cellular metabolism is rapidly converted to water. Which of the following enzymes is most likely responsible for this reaction?

- A. Glutathione peroxidase
 - B. Glutathione reductase
 - C. Superoxide dismutase
 - D. NADPH oxidase
 - E. Myeloperoxidase
 - F. Cyclooxygenase-1
-

Question #: 13

An 85-year-old man volunteers to participate in a clinical study focusing on the aging of hemopoietic organs. Imaging studies and a bone marrow biopsy are required for this study. Which of the following locations would be most appropriate for a biopsy in this patient?

- A. Iliac crest
 - B. Carpal bones
 - C. Humerus
 - D. Femur
 - E. Tibia
-

Question #: 14

A 54-year-old man comes to the physician because of a 3-month history of fatigue and weakness. Physical examination shows pale ocular and oral mucous membranes. Laboratory studies show evidence of chronic kidney disease with anemia and a low reticulocyte count. A decrease in which of the following signaling molecules most likely resulted in these laboratory findings?

- A. Interleukin 3
 - B. Interleukin 7
 - C. GM - Colony stimulating factor
 - D. Erythropoietin
 - E. Thrombopoietin
-

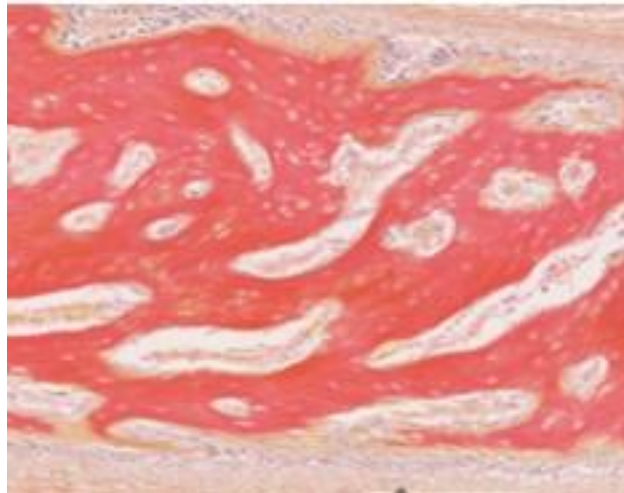
Question #: 15

A 23-year-old man comes to the physician because of a 3-day history of fever, muscle ache, headache, a cough productive of green mucus, and severe chest discomfort. His temperature is 38°C (100°F).

Physical examination shows dried oral mucosa and a "rubbing" sound at the right lung base. An x-ray of his chest shows a radiopaque collection at the right lung base. A culture of a sputum sample confirms the presence of a common flu virus. Which of the following best characterizes the prevalent cell type in this condition?

- A. Secretion of antibodies
- B. Interacts with MHC II-Ag complexes
- C. Differentiates into plasma cells
- D. Interacts with MHC I-Ag complexes
- E. Matures in the bone marrow

Attachment:



trabeculae bone (mature).jpg

Question #: 16

A 25-year-old man comes to the physician because of a 2-day history of eruption of a cluster of small, red bumps on his nose and lips. Physical examination shows a yellowish crusty sore on the nose and lips. The physician suspects a bacterial infection of his skin. Which of the following best identifies a function of the cell type most likely to be prevalent in this case?

- A. Secretion of antibodies
 - B. Secretion of interleukin 2
 - C. Secretion of interleukin 10
 - D. Secretion of TNF-alpha
 - E. Secretion of interferons
-

Question #: 17

A 67-year-old woman comes to the physician because of a 1-week history of easy bruising and gum bleeds, and a 3-month history of fatigue. Physical examination shows areas of discoloration around her body. Laboratory studies show significantly decreased platelet count. Which of the following is most likely a precursor in the development of the decreased blood element?

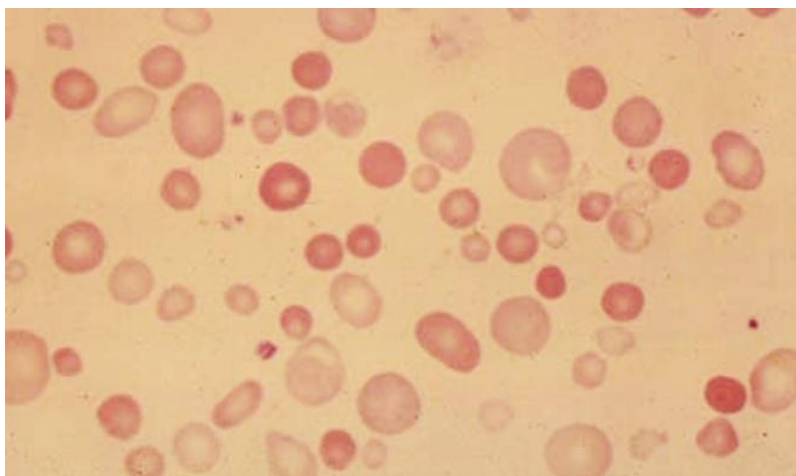
- A. Megakaryoblast
 - B. Myelocyte
 - C. Lymphoblast
 - D. Reticulocyte
 - E. Monoblast
-

Question #: 18

A 22-year-old woman comes to the physician because of a 2-month history of weakness and fatigue. Physical examination shows pallor and elevated pulse but no other abnormalities. Laboratory studies show anemia. A peripheral blood smear is shown in the attached photomicrograph. Which of the following terms best characterizes the cells of this blood smear?

- A. Anisocytosis
- B. Poikilocytosis
- C. Polycythemia
- D. Spherocytosis
- E. Target cells

Attachment:



Blood smear.jpg

Question #: 19

A 5-year-old boy is brought to the physician by his parents because of a 2-week history of abdominal pain. Stool analysis shows the presence of cysts and the ova of parasites. Laboratory studies of the blood show an increase in the count of a particular leukocyte characterized by the presence of specific granules containing major basic protein. Which of the following leukocytes are most likely elevated in this patient?

- A. Lymphocytes
 - B. Neutrophils
 - C. Monocytes
 - D. Basophils
 - E. Eosinophils
-

Question #: 20

A 25-year-old man is brought to the emergency department because of a 1-hour history of multiple injuries following a motor vehicle crash. Physical examination shows multiple, deep lacerations that require surgical intervention. Which of the following elements of the blood is required to maintain normal hemostasis?

- A. Lymphocytes
 - B. Basophils
 - C. Monocytes
 - D. Neutrophils
 - E. Thrombocytes
 - F. Eosinophils
-

Question #: 21

A 7-year-old boy is brought to the physician by his parents because of frequent respiratory infections over a period of 6 months. Physical examination shows facial and cardiac abnormalities. Karyotyping shows deletion in chromosome 22. A CT scan shows an underdeveloped thymus. Which of the following immune functions will most likely be affected?

- A. Plasma cell formation
 - B. Mast cell degranulation
 - C. Cell-mediated immunity
 - D. Phagocytosis of bacteria
 - E. Viral protein accumulation in dendritic cells
-

Question #: 22

A 22-year-old professional athlete joins a research study for the effects of endurance exercise on the body. On his first day at exercise physiology lab, he performed high-intensity exercise drills, and his parameters are noted. Which of the following cardiovascular adaptations will most likely be reported during his endurance exercise session?

- A. Decreased RV size
 - B. Decreased Diastolic filling
 - C. Decreased LV size
 - D. Unchanged LV ejection fraction
-

Question #: 23

A 28-year-old woman comes to the physician because of 3-month history of mild difficulty breathing and fatigue. She has a 10-year history of obesity grade III. Her height is 5'2" and she weighs 185 pounds, and her BMI is 35 kg/m². She is advised to start a weight loss program, including dietary modification and high intensity aerobic classes five times a week for the next six months. Which of the following effects will most likely be produced by the indicated training plan?

- A. Decreased stroke volume
 - B. Increased resting heart rate
 - C. Increased resting arterial blood pressure
 - D. Decreased ventricular muscle mass
 - E. Increased capillary capacity for blood flow
-

Question #: 24

A 25-year-old woman is brought to the emergency department after having been found unresponsive by her roommate in the bathtub. Examination shows deep lacerations to each of her wrists, and she is pale and unconscious. Her systolic blood pressure is 90 mm Hg. An increase in which of the following would

most likely be a direct consequence of her condition?

- A. Cardiac output
- B. Atrial natriuretic peptide secretion
- C. Antidiuretic hormone (vasopressin) secretion
- D. Renal blood flow
- E. Venous compliance

Question #: 25

A 62-year-old man is brought to the intensive care unit because of overwhelming sepsis. The patient has received 4 L of normal saline bolus fluids. Empirical antibiotics were begun with no improvement in his condition. His blood pressure is 60/30 mm Hg, pulse is 112/min, temperature is 40.6° C (105° F), and respiratory rate is 23/min. Which of the changes listed in the attached table are most likely in this patient before he received treatment?

- A. A
- B. B
- C. C
- D. D
- E. E
- F. F
- G. G

Attachment:

Physiologic variable	Preload (R)	Preload (L)	Pump function	Afterload	Tissue Perfusion	Treatment
Clinical Measurement	RAP/CVP	PCWP/LVEDP	Cardiac output/index	SVR/ TPR	MvO ₂	
A	↓	↓↓	↓	↑	↓	IV fluids
B	↑	↑	↓	↑	↓	Inotropes, diuresis
C	↑	↓	↓	↑	↓	Relieve obstruction
D	↑	↑	↓	↑	↓	Relieve obstruction
E	↓	↓	↑	↓	↑	IV Fluids, Pressors
F	↓	↓	↑	↓	↑	IV Fluids, Pressors
G	↓	↓	↓	↓	↑	IV Fluids, Pressors

SHOCK TABLE.jpg

Question #: 26

A 70-year-old man comes to the physician because of shortness of breath, severe fatigue and swelling of ankles for the past 3 months. Past medical history shows several anginal attacks. His vital signs are arterial blood pressure is 185/95mmHg, heart rate is 70/minute, and respirations are 17/minute. Physical examination shows mild cyanosis. Chest x-ray shows cardiomegaly and ECG shows left axis deviation. He is prescribed a drug which causes a reduction in aortic systolic pressure. Which one of the following mechanisms of action would this drug use?

- A. a decrease in arterial compliance
 - B. a decrease in venous compliance
 - C. a decrease in blood volume
 - D. an increase in systemic vascular resistance
 - E. an increase in myocardial contractility
-

Question #: 27

A 35-year-old man comes to the primary care physician with a 2-week history of progressive shortness of breath that occurs with activity. He previously exercised regularly and has never had symptoms like this before. Now he can walk only one block before becoming symptomatic. He is a non-smoker with no family history of cardiac disease. He recalls having an upper respiratory infection about a month ago that improved on its own. Physical examination shows crackles in his lungs bilaterally and an S3 gallop. An x-ray of the chest reveals cardiomegaly. He is prescribed a diuretic. What is the most likely mechanism of action of drug given to this patient for their heart failure?

- A. Increased arterial blood pressure
 - B. Increased afterload
 - C. Increased Na^+ excretion
 - D. Increased sympathetic outflow
 - E. Increased preload
-

Question #: 28

A 58-year-old man comes to the physician because of a 3-day history of severe headaches. He is hypertensive with hyperlipidemia. His blood pressure is 172/108 mm Hg but is reduced back down to normal after taking his medication. Administration of this drug is most likely to cause which of the following hormonal changes?

- A. Increased level of plasma renin
 - B. Increased levels of aldosterone
 - C. Decreased formation of angiotensin II
 - D. Decreased level of atrial natriuretic peptide
 - E. Increased levels of antidiuretic hormone
-

Question #: 29

A 25-year-old man is brought to the emergency department after suffering a head trauma in a car crash. His pulse is 57/min, blood pressure is 160/98 mmHg, and respirations are 18/min. Neurologic examination shows left hemiparesis and right anisocoria. CT scan of the head shows a frontal epidural hematoma. During the neurosurgical procedure to remove the hematoma, a catheter was placed in the left ventricle to measure intracranial pressure and estimate cerebral perfusion pressure and cerebral blood flow. The physician noticed many changes in the perfusion pressure in the range of 50 to 125 mmHg. However, cerebral blood flow did not change. Which of the following mechanisms best explains the constant cerebral blood flow in this patient?

- A. Arterial constriction
 - B. Venoconstriction
 - C. Reactive hyperemia
 - D. Metabolic hyperemia
 - E. Myogenic response
-

Question #: 30

A 5-year-old girl is brought to the emergency room after falling through the ice while skating on an outdoor pond. She is removed from the water within 1 minute, but dry clothing was not available, and she is still cold and wet 20 minutes later. Which of the following mechanisms most likely helped to maintain the patient's core temperature during the period following her rescue?

- A. Cutaneous vasodilation
- B. Diving response
- C. Increased thermoregulatory set point
- D. Release of endogenous pyrogen
- E. Shivering